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SUPERSEDING
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MILITARY SPECIFICATION

CORES FOR SEARCH COIL ASSEMBLIES

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the procurement of cores and caps for use with Search Coil Assemblies.

1.2 Classification. The material of the items shall be of the following chemical compositions, as specified (see 6.2):

Composition - Permalloy
Composition - MuMetal

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification, to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-163	Preparation of Steel Products for Domestic Shipment (Storage) and Overseas Shipment

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

FSC 1350

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3. REQUIREMENTS

3.1 General Requirements.

3.1.1 Search Coil cores and caps covered by this specification shall be manufactured in accordance with applicable Bureau of Naval Weapons Lists of Drawings.

3.1.2 Preproduction sample. Unless otherwise specified in the contract or order, a preproduction sample shall be manufactured by the same procedures and processes and at the same location as proposed for full production. This sample shall be tested for the purpose of determining compliance with the requirements of this specification, the drawings, and the contract. See 4.1.1

3.1.3 Interchangeability. Component parts of assembled units (unless otherwise specified on the applicable drawings) fabricated in strict agreement with their applicable drawings shall be interchangeable and shall assemble with mating components of the unit without selection or fitting. If, during assembly of the fabricated parts of a unit, selection or fitting is required and changes in dimensions are necessary, no changes in the parts as delineated on applicable drawings shall be made without prior approval of the Bureau of Naval Weapons.

3.1.4 Requests for deviation. Requests for deviation from this specification, applicable drawings, specifications, publications, materials, and processes specified herein, shall be referred in writing to the procuring agency.

3.1.5 Conflicting requirements. Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein, shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.2 Performance requirements and product characteristics. Cores covered by this specification shall meet the following performance requirements and product characteristics.

3.2.1 Material composition. Any composition which will meet the requirements of this specification may be used. (see paragraph 6.2).

3.2.2 Permeability.

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3.2.2.1 Raw material. The true permeability for raw material covered by this specification shall be at least 10,000 when tested as specified in 4.4.3.1.

3.2.2.2 Finished cores. The apparent initial permeability of finished cores shall be at least 95 percent of the value given by the curve of figure 1 when tested as specified in 4.4.3.2. This tolerance shall include any error due to end effect in the solenoid or threads on the ends of the core.

3.2.2.3 Caps. The average permeability of the caps shall be at least 9,500 when measured by reversal of an average magnetizing force of H = .002 oersted as specified in 4.4.3.3. Average permeability shall be defined as the ratio of the average flux density, B, to the average magnetizing force, H. Samples from each heat shall be tested.

3.2.3 Straightness. Each finished core shall pass the straightness test as specified in 4.4.4.

3.3 Workmanship and cleanliness. The core material and completed cores and caps shall be free from dirt, metal chips, filings and other foreign matter. Since mechanical stresses and strains are apt to impair the permeability of the core material, extreme care shall be exercised in handling. The contractor shall be responsible for the proper handling of core material and of the finished cores to insure that the usefulness of the product has not been impaired.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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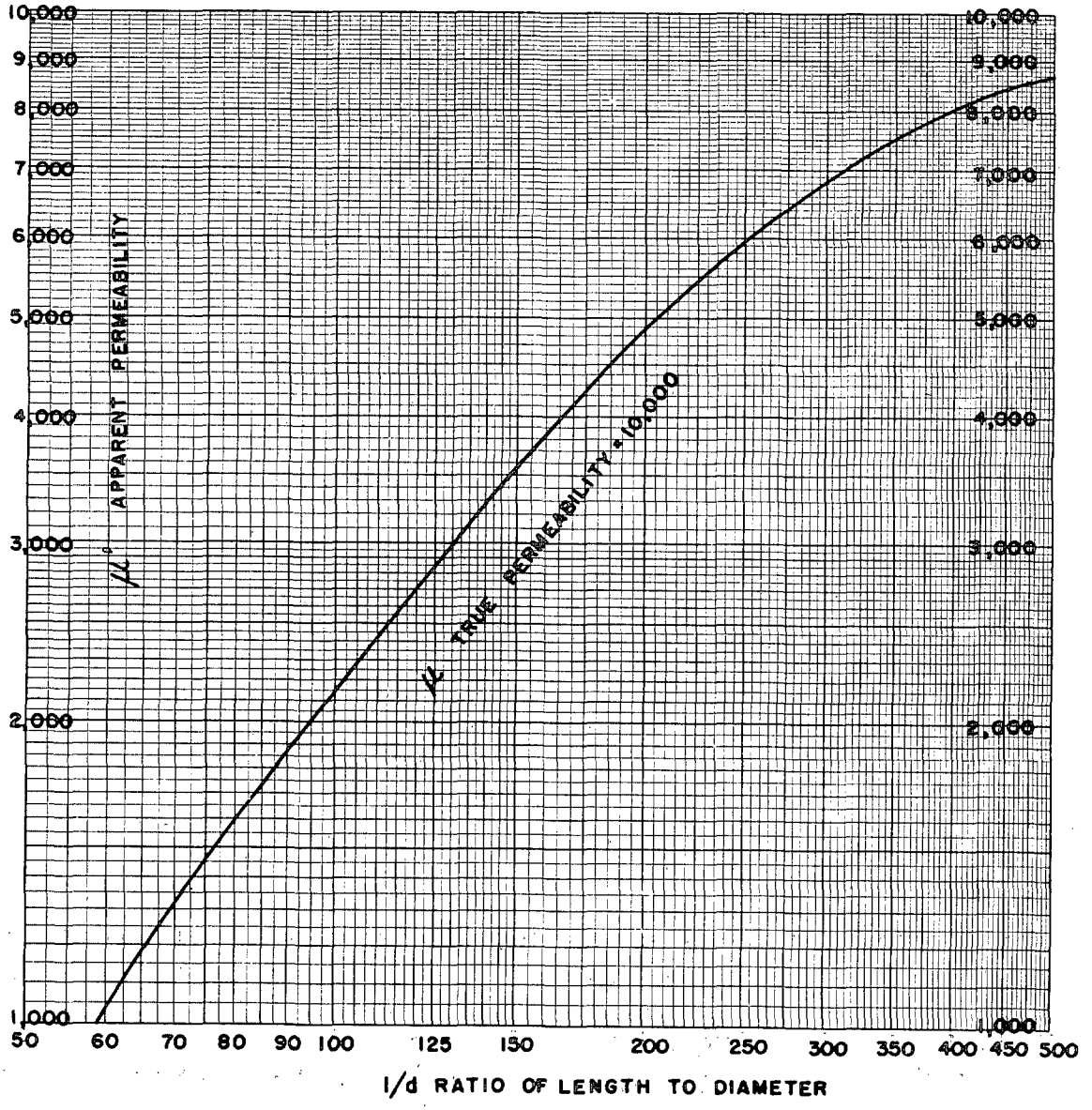


FIGURE 1

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4.1.1 Preproduction sample. Unless otherwise specified for each contract or order, a representative sample of 10 cores shall be manufactured using the methods and procedures proposed for full production. The preproduction sample shall be forwarded to an activity designated in the contract or by the procuring agency, for the purpose of determining compliance with the requirements of the contract, specifications and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of the tests on preproduction samples shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order.

4.1.1.1 Preproduction sample for a subsequent contract. The necessity for a preproduction sample will be determined by the procuring agency when production under a new contract by the same contractor at the same location follows the manufacture of satisfactory material as covered by this document.

4.2 Classification of tests. The inspection and testing of cores covered by this document shall be classified as follows:

(a) Preproduction tests
(b) acceptance tests

4.2.1 Preproduction tests. Preproduction tests are those performed on samples submitted in accordance with 4.1.1 to determine that the lot and production meet all the requirements of this specification. Acceptance shall be based on no defects in a sample.

4.2.2 Acceptance tests. Acceptance tests for acceptance of the material covered by this specification shall consist of tests to determine compliance with all requirements specified herein.

4.3 Test equipment.

4.3.1 Test and inspection equipment and facilities. Unless otherwise specified, the manufacturer or commercial laboratory (see 4.1) shall furnish and maintain all necessary test equipment, facilities, and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity, and unless definite requirements are specified, they shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

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4.3.1.1 Equipment required for performing these tests is as follows:

(a) Search Coil Test Solenoid (LD 295670) (Government will furnish)
(b) Fluxmeter, Recording Mark 2 Mod 3 or 4 (Government will furnish)
(c) Transformer Mark 17 Mod 0 or 1 (Government will furnish)
(d) Fluxmeter Control Box - Navy Stock Number Z1075-334-3602 (Government will furnish)
(e) Fixed Resistors:
One each (R1) 10 ohms, 3%, 5 watt
(R2) 100 ohms, 3%, 5 watt
(f) Rheostats:
One each (R3) 0-5 ohms, 3 turns, 3 watt
(R4) 0-2000 ohms, 10 turns, 3 watt
(g) Milliammeters:
One each (M1) capable of reading 450 ± 2 milliamperes(ma), calibrated at 450 ma.
One each (M2) having two ranges with the following capabilities:

Low Range - 0-10 ma, accuracy ± 0.05 ma, calibrated at 5.00 ma.
High Range - 10-100 ma, accuracy ± 0.5 ma, calibrated at 30.0 ma.

(h) Single Throw Switch
One each (S1) - must carry 5 amperes inductive current at 6 volts; contact resistance not greater than 0.01 ohm.

(i) Current Source - One 6-volt storage battery

4.4 Tests

4.4.1 Test conditions. Unless otherwise specified, tests shall be performed at 80 ± 10 degrees(°) Fahrenheit (F) and ambient relative humidity.

4.4.2 Visual inspection. The cores and caps shall be inspected to verify that the materials, design, construction, physical dimensions, and workmanship are in accordance with this document and the drawings.

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4.4.3 Permeability tests.

4.4.3.1 Raw material test. To determine compliance with the requirements of 3.2.2.1, a sample of the core material shall be taken from each melt and subjected to substantially the same heat treatment as the material for SC assemblies. The true permeability of this sample, measured at a magnetizing force of H = 0.002 oersteds, shall be at least 10,000. The type of sample and the method of measurement shall be approved by the Bureau of Naval Weapons.

4.4.3.2 Finished core permeability test. To determine compliance with the requirements of 3.2.2.2, each core shall be subjected to the following test

(a) The search coil test solenoid, test item 4.3.1.1 (a), shall be placed in an east-west direction to eliminate the background field. This direction shall be determined with a compass and shall not vary over the length of the solenoid.

(b) The transformer, test item 4.3.1.1 (c), shall be placed at least three feet from any iron object, and far enough from the solenoid to prevent inter-action. The required distance from the solenoid is about ten feet, with the axis of the transformer perpendicular to the solenoid axis. The test for interaction shall be made by connecting the fluxmeter, test item 4.3.1.1 (b), to the solenoid secondary and a 6-volt storage battery, test item 4.3.1.1 (i), to the transformer primary. With a core of the type being tested in place in the solenoid, a change of 500 milliamperes current in the transformer primary should give no detectable fluxmeter response.

(c) As shown in the schematic diagram of figure 2, one side of the primaries of the solenoid and the transformer shall each be connected in series with a fixed resistor, a rheostat, and a milliammeter. The milliammeters shall both then be connected to one side of S1, the single-throw switch, test item 4.3.1.1 (h). The other sides of the two primaries shall each be connected to the positive terminal of the 6-volt storage

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battery, test item 4.3.1.1 (i). The negative terminal of the storage battery shall be connected to the other terminal of switch S1. This single-throw switch will make and break the two primary currents simultaneously. This circuit has been designed to allow the control and measurement of current in each of the primaries.

(d) The secondaries of the solenoid and the transformer shall be connected in series to the recording fluxmeter through the fluxmeter control box. This connection shall be made in such a way that when the primary circuit is made, the transformer tends to give a fluxmeter deflection to the left and the solenoid tends to give an opposite fluxmeter deflection to the right. Correct polarity can be established by energizing each primary separately, in turn.

(e) In the test of a core, the current in the primary of the transformer shall be adjusted to 450 milliamperes by varying R3 and reading at meter M1. Unless otherwise specified, the current in the solenoid primary is calculated from:

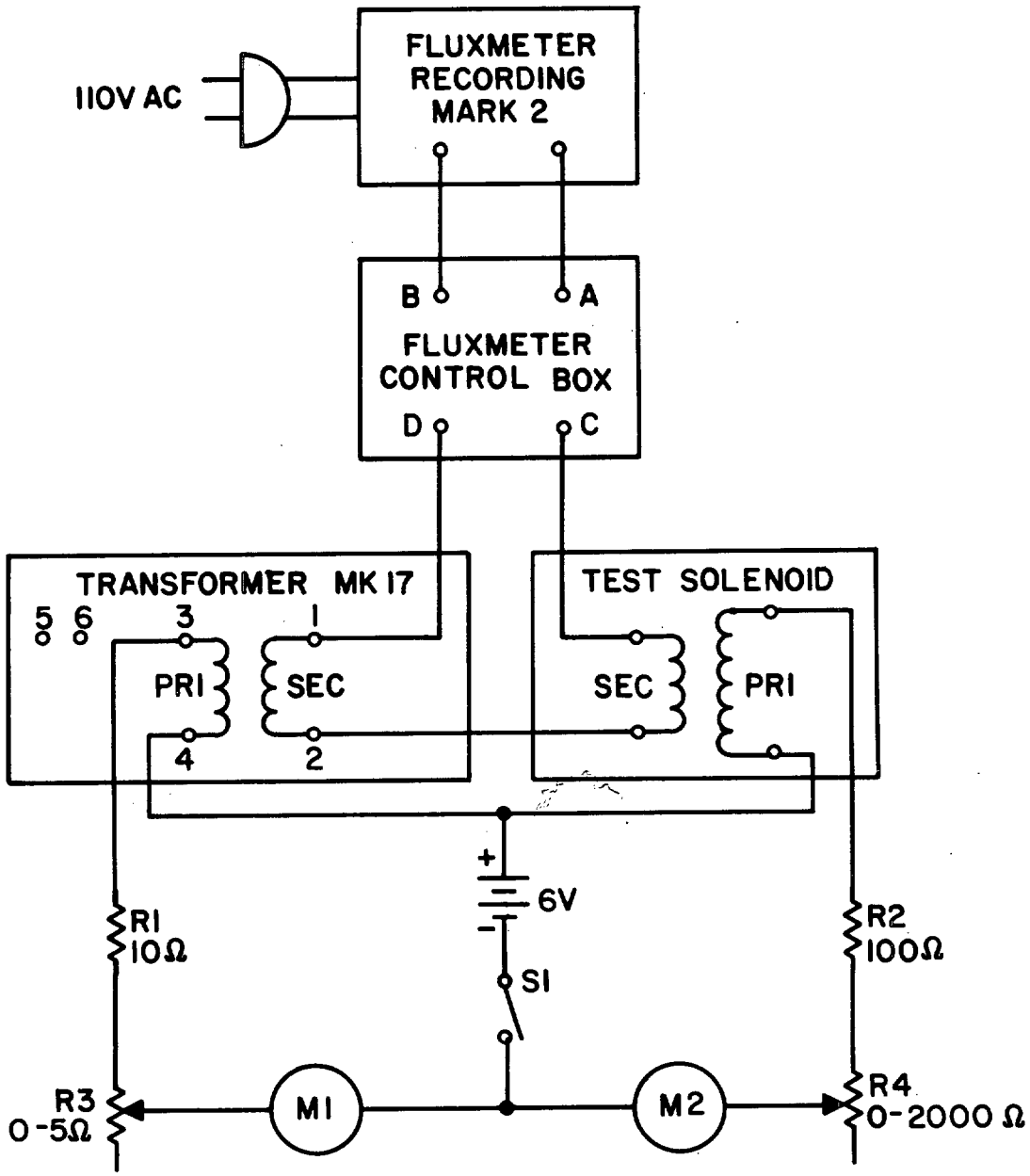
$$\frac{23,700}{d^2 \mu'} \text{ milliamperes}$$

Where: d is the nominal core diameter in inches

and μ' is the value of the apparent permeability given by the curve of figure 1 for the nominal value of length/diameter ratio of the core.

Adjust R4 to obtain this current reading on meter M2. Before proceeding, check to see that both M1 and M2 are reading as required. R3 and R4 shall be adjusted, in turn, until both meters read correctly.

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CIRCUIT FOR PERMEABILITY TEST OF CORES

FIGURE 2

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(f) The core shall be centered longitudinally and coaxially in the solenoid and must be continuously supported along its entire length.

(g) With the current off, center the fluxmeter pointer and adjust to zero drift with the use of the Fluxmeter Control. Leave the pointer centered.

(h) Make the current for a period of a few seconds. Hold sufficiently long for the fluxmeter pointer to settle down to a steady position. Break the current.

(i) If the steady position of the pointer, while the current is on, is to the left of the steady position with the current off, the core has failed to meet the requirement 3.2.2.2. If the position, with current on, is to the right of, or coincides with, the position with current off, the core is acceptable. (The 95 percent factor has already been allowed for in the current values specified in step (e) above.)

4.4.3.3 Caps permeability test. To determine compliance with the requirements of 3.2.2.3, three randomly selected samples from each heat shall be tested as ring samples. Failure of the sample caps to meet requirement 3.2.2.3 shall result in the rejection of that heat. A primary and a secondary coil shall be wound on the ring and the flux linkage in the secondary measured for reversal of a given current in the primary.

H, the average magnetizing force, shall be determined from:

$$H = \frac{0.4\ NpIp\ \log_e\ (d_2/d_1),}{d_2 - d_1}$$

where

N_p is the number of turns on the primary coil,

I_p is the primary current in amperes (to be chosen so that H has the value of 0.002),

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d_2 is the outside diameter of the cap in centimeters,

d_1 is the inside diameter of the cap in centimeters (at the root of the thread).

B, the average flux density, shall be determined from:

$$B = L/2AN_S$$

where

$A = 1/2\ (d_2 - d_1)t$,

L is the flux linkage change in the secondary coil which occurs upon reversal of primary current I_p,

N_S is the number of turns on the secondary coil,

t is the thickness of the cap in centimeters.

4.4.4 Straightness test. To determine compliance with the requirements of 3.2.3, each finished core shall pass freely through a tube of the same length as the core, with an inside diameter 1/16 inch greater than the outside diameter of the core.

4.5 Acceptance criteria. Failure of any core or cap to meet any of the requirements and tests of this specification shall result in rejection of the core or cap. If the number of failing cores or caps exceeds the number permitted by the associated criteria, the lot represented will be rejected.

4.6 Preservation, packing and packaging. Preservation, packing and packaging of the cores shall conform to section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. (Not applicable)

5.1.2 Level B.

5.1.2.1 Cleaning and drying. (Not applicable, see 3.3)

5.1.2.2 Preservative application. The selection of preservatives shall be in accordance with requirements of Military Standard MIL-STD-163, as required.

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5.1.2.3 Unit packaging. Subject material shall be packaged as specified in accordance with the requirements of Military Standard MIL-STD-163.

5.1.3 Level C.

5.1.3.1 Cleaning and drying. (Not applicable, see 3.3)

5.1.3.2 Preservative application. The selection of preservatives shall be in accordance with requirements of Military Standard MIL-STD-163, as required.

5.1.3.3 Unit packaging. Subject material shall be packaged as specified in accordance with the requirements of Military Standard MIL-STD-163.

5.2 Packing.

5.2.1 Level A. (Not applicable)

5.2.2 Level B.

5.2.2.1 Exterior containers. Subject material, packaged as described in 5.1.2, shall be packed in accordance with the requirements of Military Standard MIL-STD-163.

5.2.3 Level C.

5.2.3.1 Exterior containers. Subject material packaged as described in 5.1.3, shall be packed in accordance with the requirements of Military Standard MIL-STD-163.

5.3 Marking. In addition to any special marking required by the contract or order, all marking shall be in accordance with Military Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the title, number, and date of this specification, and exceptions to this specification, applicable drawings and other documents. Procurement documents should also specify the desired level of packaging and packing. The attention of the contracting officer is invited to the Quality Assurance Provisions and options in Specification MIL-P-116, and to paragraph 6.1.1 of this document.

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6.1.1 Criteria for use of proper level of preservation, packaging and packing shall be as follows:

For Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For Level B. This level shall be used only when it is definitely known that the item will be held in covered storage in overseas locations for 6 months or less or in domestic locations for an indefinite period.

For Level C. This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

6.2 Material. The following compositions have been tested and approved. Any other product offered as an alternate must be tested and approved by the Bureau of Naval Weapons prior to use in this application.

(a) Permalloy	Chemical comp. (approx)
Nickel	79 percent
Iron	17 percent
Molybdenum	4 percent

(b) Mu Metal	Chemical comp. (approx)
Nickel	76 percent
Iron	16 percent
Copper	6 percent
Chromium	2 percent

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